

ONLINE TRAVEL BOOKING SYSTEM



Project Proposal

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CERTIFICATION

This document has been written by the project group under the directions of our supervisor and co-supervisor and is submitted as a Final Year Project proposal for academic evaluation. The project — Online Travel Booking System — is a comprehensive PHP/MySQL-based web platform designed to automate and streamline travel bookings for customers and administrators.

DECLARATION

We hereby declare that we completely write this document. It represents our original effort and no person outside our group has contributed to its content. This proposal is written in a technical manner in accordance with our project requirements.

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We are grateful to our teachers, friends, and family whose continuous support enabled us to successfully undertake this work. We are also thankful to the staff of National Textile University, Faisalabad for providing services whenever needed.

ABSTRACT

This project presents the design and development of an Online Travel Booking System — a fully integrated web-based platform built using PHP, MySQL, and modern CSS/JavaScript technologies. The system automates the complete lifecycle of travel bookings, from package browsing and trip customization to payment processing and administrative oversight, replacing fragmented manual workflows with a centralized digital solution.

The system implements a role-based architecture with three distinct user levels: Guests, Registered Users, and Administrators. Guests can browse travel packages and register for accounts. Registered users can submit detailed booking requests, make payments, and track their booking history. Administrators have full control over packages, bookings, payments, and users through a dedicated admin panel, with real-time dashboard statistics for informed decision-making.

Key features include a hero slider homepage with dynamic package listings, a multi-step booking form capturing 14 travel parameters, an integrated payment gateway, an employee management portal, and a comprehensive admin dashboard. The system ensures data security through session-based authentication, password hashing, and input sanitization, providing a secure, scalable, and user-friendly travel management experience.

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CHAPTER 1

1 PROJECT INITIATION

1.1 Introduction

The Online Travel Booking System (OTBS) is a modern, web-based platform developed to fully automate and streamline the travel booking process for both customers and administrators. Built using PHP, MySQL, HTML5, CSS3, and JavaScript, the system provides a comprehensive digital environment where users can explore travel destinations, select from a variety of curated packages, submit detailed booking requests, and complete secure online payments — all through a single, unified interface.

The platform encompasses a public-facing website (home.php, package.php, book.php, payment.php), a user portal for managing personal bookings (my_bookings.php), and a full-featured administrative control panel (admin/dashboard.php) for managing bookings, packages, users, and payments. An additional employee portal (employee.php) enables HR and management tasks including payroll tracking, leave management, and performance monitoring.

1.2 Background of the Study

The global travel industry has undergone a significant digital transformation over the past decade. Traditional travel agencies relying on manual processes, paper-based record keeping, and telephone-based booking have increasingly been replaced by online platforms that offer instant booking confirmation, real-time availability, and digital payment processing.

In Pakistan's growing travel and tourism sector, many small-to-medium travel businesses still rely on fragmented manual workflows: maintaining bookings in spreadsheets, accepting payments via bank transfers with manual reconciliation, and communicating with clients through WhatsApp or phone. This creates inefficiencies including double bookings, payment tracking errors, and poor customer experience.

The Online Travel Booking System addresses these challenges by providing a centralized, automated digital platform tailored to the needs of modern travel businesses, enabling them to compete effectively in an increasingly digital marketplace.

1.3 Problem Statement

Current travel booking processes for small and medium travel agencies in Pakistan suffer from several critical deficiencies:

- Manual booking records are error-prone, leading to double bookings and data loss.
- No centralized system exists to track booking status (pending, confirmed, cancelled) in real time.
- Payment collection is disorganized with no automated recording or receipt generation.
- Customers lack a self-service portal to view their booking history or track payment status.
- Administrative tasks (managing users, packages, payments) are time-consuming without proper tooling.
- There is no integrated employee management system, leaving HR tasks fragmented.

The Online Travel Booking System solves all these problems by providing a secure, role-based, database-driven platform that automates the complete travel booking lifecycle.

1.4 Project Scope

The scope of the Online Travel Booking System defines the functional boundaries, system capabilities, and user interaction layers of the project.

1.5 Public Website & Package Discovery

The public-facing website (home.php) features a full-screen hero slider highlighting destinations, a statistics strip displaying total users and bookings, and a curated featured packages section. The packages page (package.php) provides filtering by destination type (Adventure, Honeymoon, Luxury, Family, Beach) with dynamic pricing display. Visitors can browse packages without registration but must create an account to book.

1.6 User Registration & Authentication

The system includes a full authentication module with separate login portals for customers (login.php) and administrators (admin/login.php). Registration (register.php) collects name, email, phone, and password with real-time password strength checking. Both portals implement session-based authentication with PHP session management, and passwords are stored as bcrypt hashes for security.

1.7 Booking Management System

The booking form (book.php / book_form.php) captures 14 distinct travel parameters including: full name, email, phone, address, destination (location), number of guests, arrival date, departure date, trip type, accommodation preference, meal preference, transport preference, budget per person per night, and special requests. The system automatically calculates the total amount based on guests, nights, and budget. Bookings are stored with a 'pending' status and the user is redirected to the payment page immediately after submission.

1.8 Payment Processing Module

The payment module (payment.php) supports multiple payment methods including credit/debit card and bank transfer. Card payments include validation of card number (13-19 digits), cardholder name, expiry date, and CVV. Only the last 4 digits of the card number are stored in the database for security. Upon successful payment submission, the booking status is updated, and a payment record is created in the payments table with 'pending' status for admin verification.

1.9 User Booking Portal

Registered users can access their personal bookings dashboard (my_bookings.php) showing all past and current bookings with status indicators (pending, confirmed, cancelled), booking statistics (total, confirmed, pending, cancelled), and detailed booking information including destination, dates, guests, payment amount, and trip type.

1.10 Admin Control Panel

The administrator panel (admin/) provides complete oversight and control of the system through six modules: Dashboard (real-time stats), Bookings Management (view/update/delete all

bookings), Packages Management (add/edit/delete travel packages), Payments Management (verify and manage payment records), and Users Management (view/delete registered users). The dashboard displays six key metrics: total bookings, pending bookings, confirmed bookings, cancelled bookings, total users, and total packages.

1.11 Employee Portal

An integrated employee management system (employee.php) tracks 10 staff members across 8 departments including Tour Operations, Customer Service, Finance, Marketing, IT & Systems, HR, Guides, and Logistics. Features include employee directory with performance scores and status indicators (active, remote, on leave, inactive), payroll statistics, leave request management with approval/rejection workflow, and departmental distribution reports.

1.12 Stakeholders

Stakeholder	Role	Interest	Importance
Developers	Technical Architects	System performance, security, database integrity, PHP/MySQL stack	High
Administrators	System Operators	Booking management, package control, user oversight, payment verification	High
Registered Users	Customers	Easy booking, payment processing, booking history access	High
Guests	Visitors	Package browsing, destination discovery, registration	Medium
Employees	Staff	HR portal access, leave management, payroll information	Medium

CHAPTER 2

2 LITERATURE REVIEW & TECHNOLOGY STACK

2.1 Existing Systems Review

Several commercial travel booking platforms exist globally, including Booking.com, Expedia, and Kayak. These platforms use enterprise-grade architectures with microservices, third-party API integrations (airline GDS, hotel inventory systems), and complex payment gateways. While feature-rich, they are beyond the scope and budget of small-to-medium Pakistani travel agencies.

Locally, most travel agencies rely on manual processes or generic e-commerce platforms not tailored to travel. Some use simple HTML contact forms that submit booking requests via email, with no database storage, status tracking, or payment processing. The Online Travel Booking System fills this gap by providing a purpose-built, database-driven solution with all necessary features for a fully operational travel business.

2.2 Technology Stack

Component	Technology	Purpose
Backend Language	PHP 8.x	Server-side scripting, session management, database operations, form processing
Database	MySQL (booking_db)	Relational data storage for users, bookings, packages, and payments
Database API	MySQLi (Procedural)	Secure database queries with prepared statements for injection prevention
Frontend	HTML5 / CSS3 / JavaScript	Responsive UI, form validation, interactive components, hero sliders
CSS Framework	Custom + Tailwind utilities	Responsive design, consistent styling across all pages
Icon Library	Font Awesome 6.4	UI icons throughout the application
Typography	Google Fonts (Playfair Display, DM Sans)	Professional typographic hierarchy
Slider	Swiper.js 7	Hero image carousel on homepage
Authentication	PHP Sessions + bcrypt	Secure session-based auth with hashed passwords
Server	Apache / XAMPP (localhost)	Local development and production deployment

2.3 PHP vs. Framework-Based Approaches

The system uses native PHP with MySQLi rather than a framework such as Laravel or CodeIgniter. This decision was made for the following reasons:

- **Simplicity:** Native PHP provides complete control without the overhead of learning framework conventions, making development faster for this project scope.
- **Portability:** The system runs on any standard LAMP/WAMP/XAMPP stack without Composer dependencies or complex configuration.
- **Educational value:** Native PHP exposes the full request-response cycle, database operations, and session management, which is appropriate for an academic project demonstrating core web development competencies.
- **Performance:** For the scale of this application, native PHP performs comparably to framework-based solutions without the framework's initialization overhead.

2.4 Security Technologies

2.5 Password Hashing with bcrypt

All user and admin passwords are stored using PHP's `password_hash()` function with the `PASSWORD_DEFAULT` algorithm (bcrypt). Authentication uses `password_verify()` to compare

submitted passwords against stored hashes. This ensures that even if the database is compromised, plain-text passwords cannot be recovered.

2.6 SQL Injection Prevention

The admin login module uses `mysqli_prepare ()` and `mysqli_stmt_bind_param ()` for parameterized queries, completely preventing SQL injection attacks. The user-facing modules use `mysqli_real_escape_string ()` for input sanitization as an additional layer of protection.

2.7 Session Management

PHP sessions are used throughout the application to maintain authentication state. Every protected page check for the appropriate session variable (`$_SESSION['admin_id']` or `$_SESSION['user_id']`) at the top of the file and redirects to the login page if not set, preventing unauthorized access to protected resources.

CHAPTER 3

3 METHODOLOGY AND REQUIREMENTS

3.1 Agile Methodology

The development of the Online Travel Booking System followed the Agile Software Development Lifecycle (SDLC), organized into four two-week sprints:

- Sprint 1 — Core Infrastructure: Database schema design (`booking_db`), PHP session-based authentication for both user and admin portals, and the home page with dynamic statistics.
- Sprint 2 — Booking & Payment: Multi-parameter booking form (`book.php/book_form.php`), automatic total calculation logic, and the payment processing module (`payment.php`) with card validation.
- Sprint 3 — Admin Panel: Complete admin dashboard (`dashboard.php`), bookings management (`bookings.php`), users management (`users.php`), packages management (`packages.php`), and payments management (`payments.php`).
- Sprint 4 — User Portal & Polish: User booking history (`my_bookings.php`), employee portal (`employee.php`), support pages (FAQ, Privacy, Terms), and end-to-end testing with UI refinement.

3.2 Functional Requirements

3.2.1 Guest (Unauthenticated) Requirements

- G-1: Browse homepage with hero slider, statistics, and featured packages.
- G-2: View full packages listing with category filtering (Adventure, Honeymoon, Luxury, Family, Beach).
- G-3: View About page, FAQ page, Privacy Policy, and Terms of Use.
- G-4: Register a new account with name, email, phone, and password.
- G-5: Login to an existing account.

3.2.2 Registered User Requirements

- U-1: Submit a detailed booking request with 14 travel parameters.
- U-2: Complete payment after booking with card or bank transfer.
- U-3: View personal booking history with status tracking.
- U-4: View booking statistics (total, confirmed, pending, cancelled bookings).
- U-5: Logout securely from the session.

3.2.3 Administrator Functional Requirements

- AFR-1: Login to the dedicated admin panel with email and password.
- AFR-2: View real-time dashboard with 6 key performance metrics.
- AFR-3: View and manage all bookings — update status (pending/confirmed/cancelled), delete bookings.
- AFR-4: Full CRUD operations on travel packages including destination, duration, price, category, and description.
- AFR-5: View all payment records and manage payment statuses.
- AFR-6: View all registered users, search by name/email, view user details, delete users.
- AFR-7: View recent bookings on the dashboard for quick overview.
- AFR-8: Access to Employee Portal for HR and payroll management.

3.3 Non-Functional Requirements

3.3.1 Security

- All passwords must be hashed using PASSWORD_DEFAULT (bcrypt) before database storage.
- Admin login must use prepared statements (mysqli_prepare) to prevent SQL injection.
- All protected pages must validate session at the top of the file before rendering any content.
- Card numbers must never be stored in full — only the last 4 digits may be retained.

3.3.2 Usability

- The interface must be fully responsive, adapting to mobile (360px), tablet (768px), and desktop (1440px+) viewports.
- Form validation must provide immediate, descriptive error messages to guide users.
- The admin panel must use consistent sidebar navigation across all admin pages.

3.3.3 Performance

- Database queries must complete within 200ms for all standard operations.
- Pages must load within 3 seconds on a standard broadband connection.
- JavaScript used for UI enhancement (search filtering, modals) must not block page rendering.

3.4 Hardware and Software Requirements

Category	Requirement	Specification
Software	Operating System	Windows 10/11 or macOS
Software	Web Server	Apache 2.4+ (via XAMPP or WAMP)
Software	PHP Version	PHP 8.0 or higher
Software	Database	MySQL 8.0+ / MariaDB 10.6+
Software	Browser	Chrome 90+, Firefox 88+, Edge 90+ (latest)
Software	Development IDE	VS Code, PhpStorm, or Notepad++
Hardware	Processor	Intel Core i3 (min); i5/i7 (recommended)
Hardware	RAM	4 GB minimum; 8 GB recommended
Hardware	Storage	256 GB SSD; minimum 1 GB for project files

CHAPTER 4

4 SYSTEM ANALYSIS AND DESIGN

4.1 System Architecture

The Online Travel Booking System follows a three-tier web architecture pattern:

- Presentation Layer: HTML5/CSS3/JavaScript pages rendered by the PHP engine, delivered to the client browser. Includes both the public-facing site and the admin panel.
- Application Layer: PHP scripts on the Apache web server handle business logic, session management, form processing, authentication, and data validation.
- Data Layer: MySQL database (booking_db) stores all persistent data across five primary tables: users, bookings, packages, payments, and admin.

4.2 Database Schema (ERD)

The database (booking_db) contains five primary tables with the following structure and relationships:

Table	Primary Key	Foreign Keys	Key Fields
users	id (INT, AI)	None	name, email (UNIQUE), phone, password (hashed), created_at
admin	id (INT, AI)	None	name, email (UNIQUE), password (hashed)
packages	id (INT, AI)	None	title, destination, duration, price, category, description, image, status (active/inactive)

bookings	id (INT, AI)	user_id → users.id	name, email, phone, address, location, guests, arrival, leaving, trip_type, accommodation, meal, transport, budget, special_requests, status (pending/confirmed/cancelled), created_at
payments	id (INT, AI)	booking_id → bookings.id, user_id → users.id	amount, payment_method, card_last4, card_name, status (pending/verified/failed), created_at

4.2.1 Entity Relationship Details

- One User can have Many Bookings (1: N relationship — users.id → bookings.user_id).
- One Booking has One Payment (1:1 relationship — bookings.id → payments.booking_id).
- One User can have Many Payments (1: N relationship — users.id → payments.user_id).
- Packages are independent entities linked to Bookings conceptually through the location/destination field.
- Admin table is completely independent from Users, representing a separate authentication domain.

4.2.2 ERD Diagram Description

The Entity Relationship Diagram below illustrates the complete data model of the system. The central entity is Bookings, which connects Users (who create bookings) to Payments (which are generated per booking). Packages exist as a reference catalog. The admin entity is a separate authentication domain with no foreign key relationships to other tables.

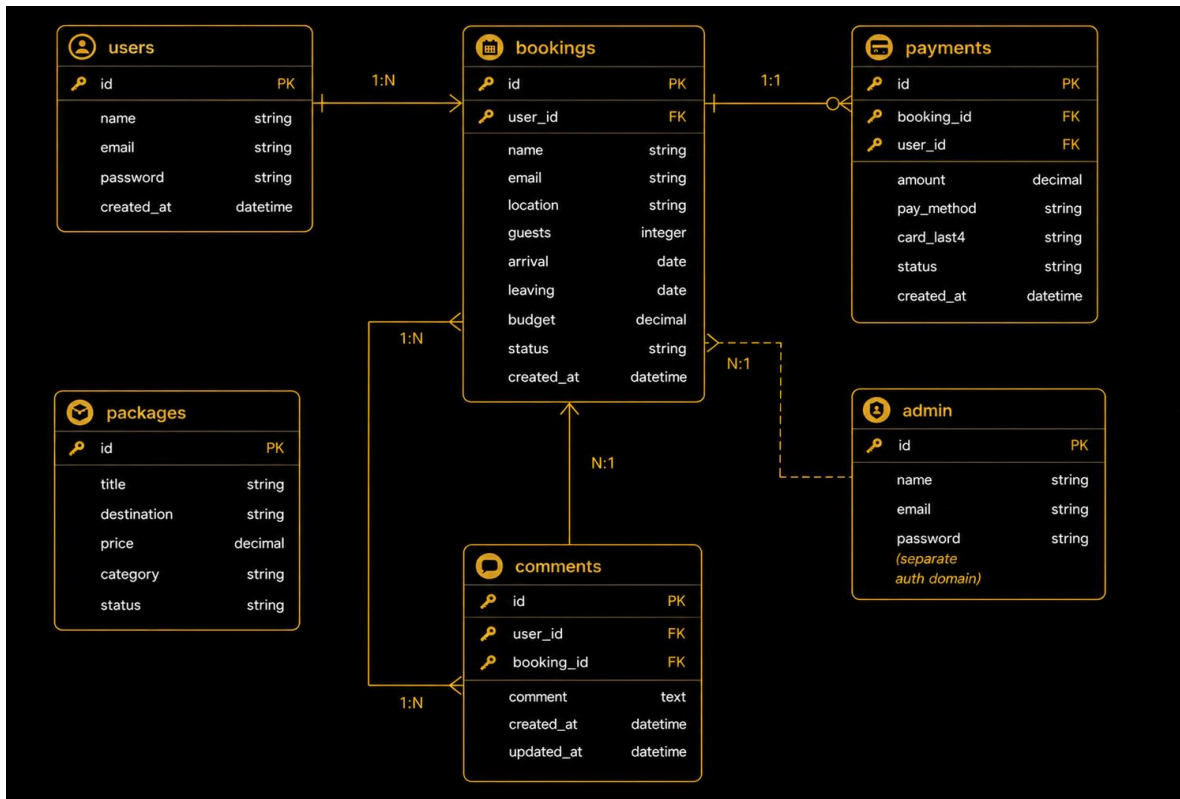


Figure 1: Entity Relationship Diagram (ERD)

4.2.3 Use Case Diagram

The Use Case Diagram identifies three primary actors — Guest, Registered User, and Administrator — and maps their interactions with the system's functional modules.

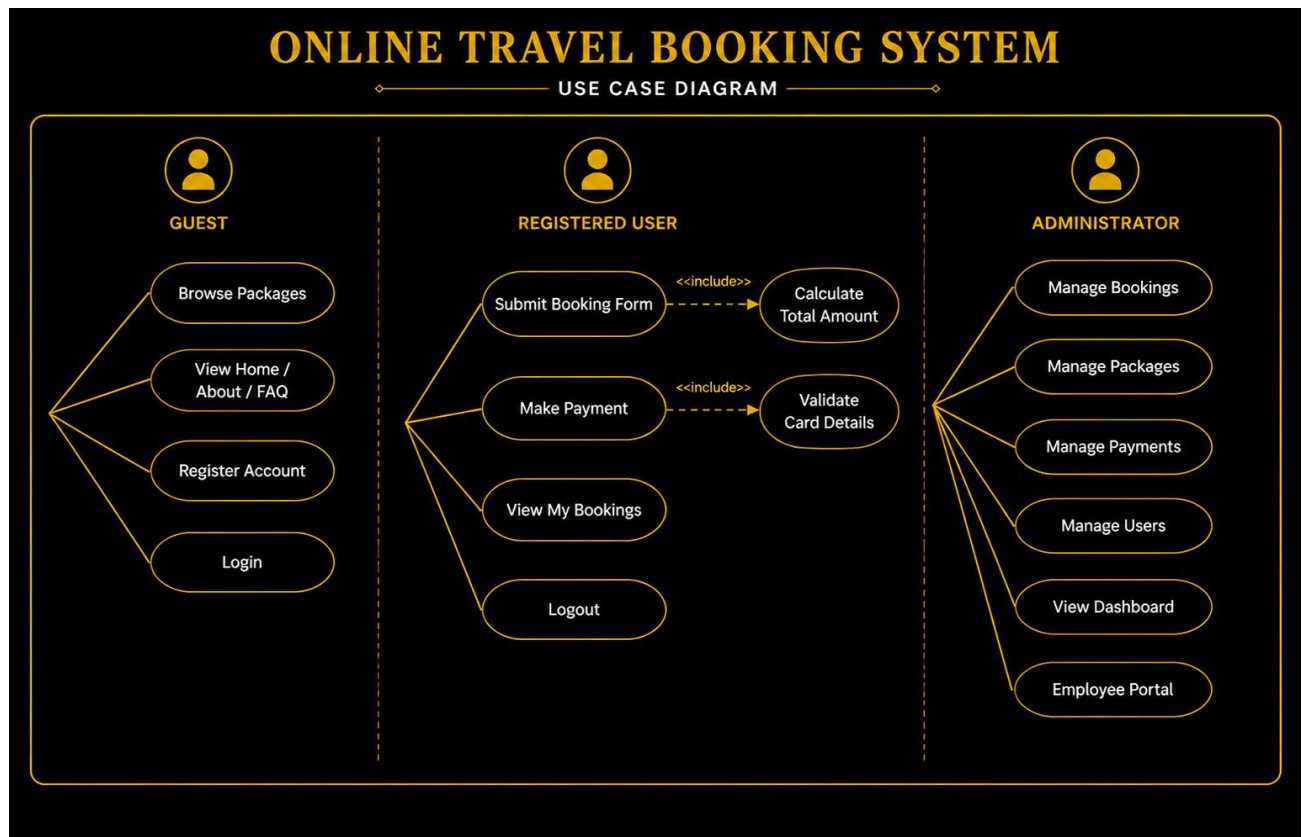


Figure 2: Use Case Diagram

4.3 4.4 Activity Diagram

The following Activity Diagram illustrates the complete booking flow from user arrival on the website through to successful payment confirmation.

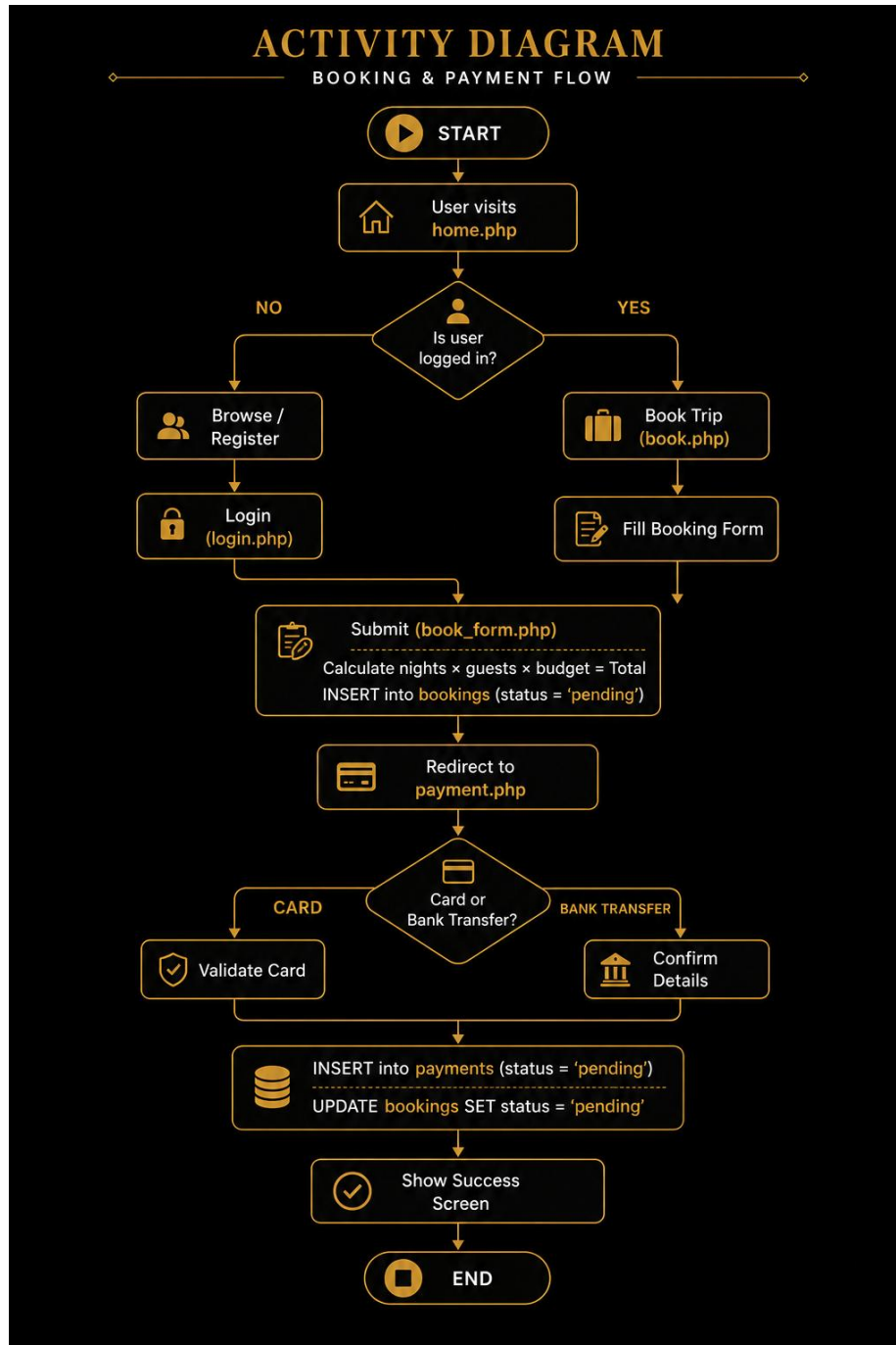


Figure 3: Activity Diagram — Booking & Payment Flow

4.3.1 Sequence Diagram

The Sequence Diagram illustrates the interaction between system components during the booking process, showing the temporal order of messages between the User, Browser, PHP Server, and MySQL Database.

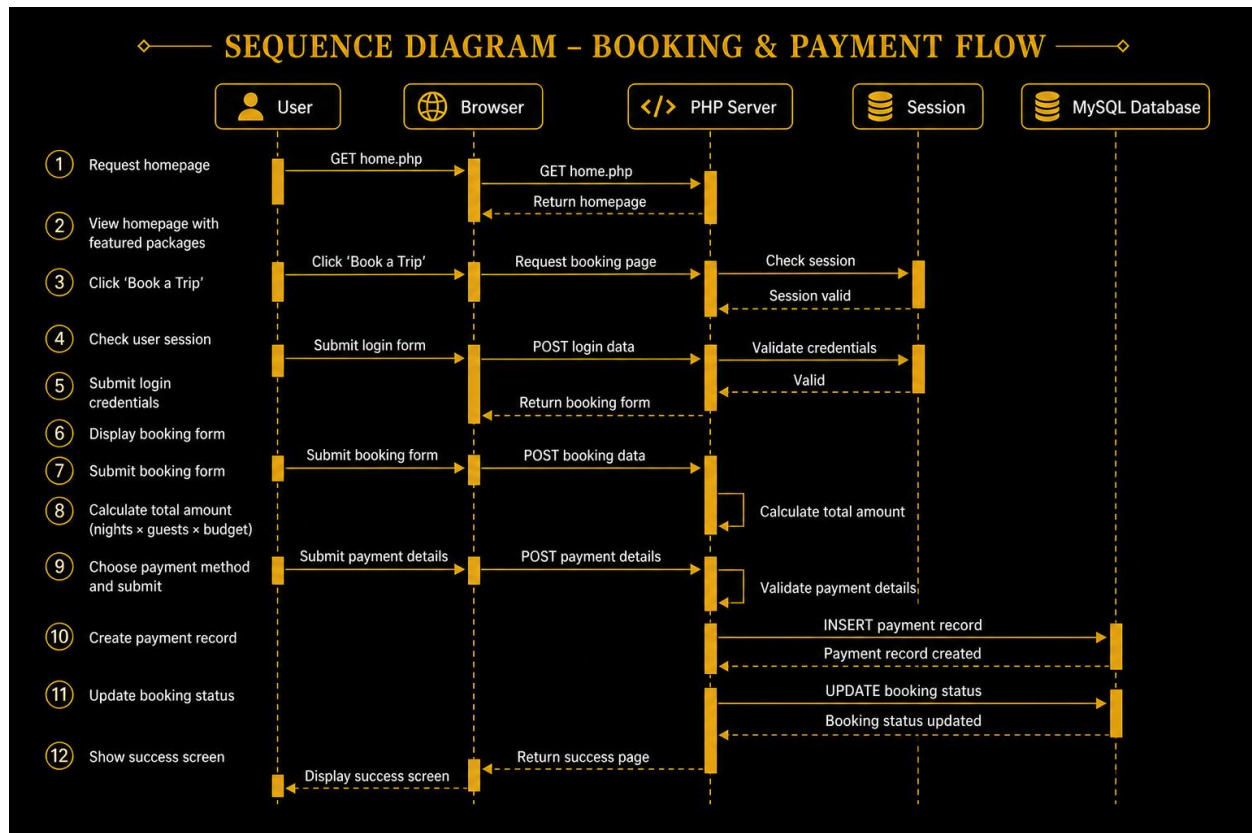


Figure 4: Sequence Diagram — Complete Booking Flow

4.3.2 System Flowchart

The system flowchart presents the decision logic for the two primary system portals: the user-facing booking portal and the administrator management panel.

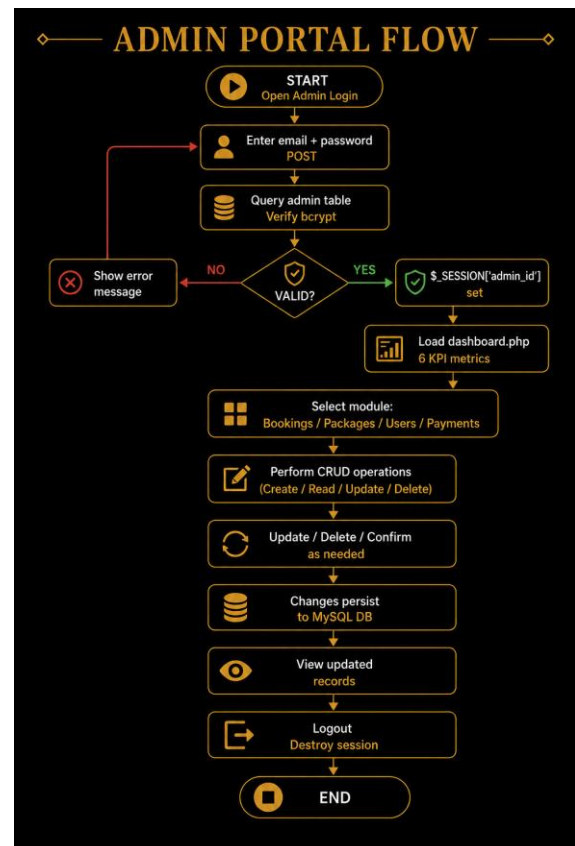
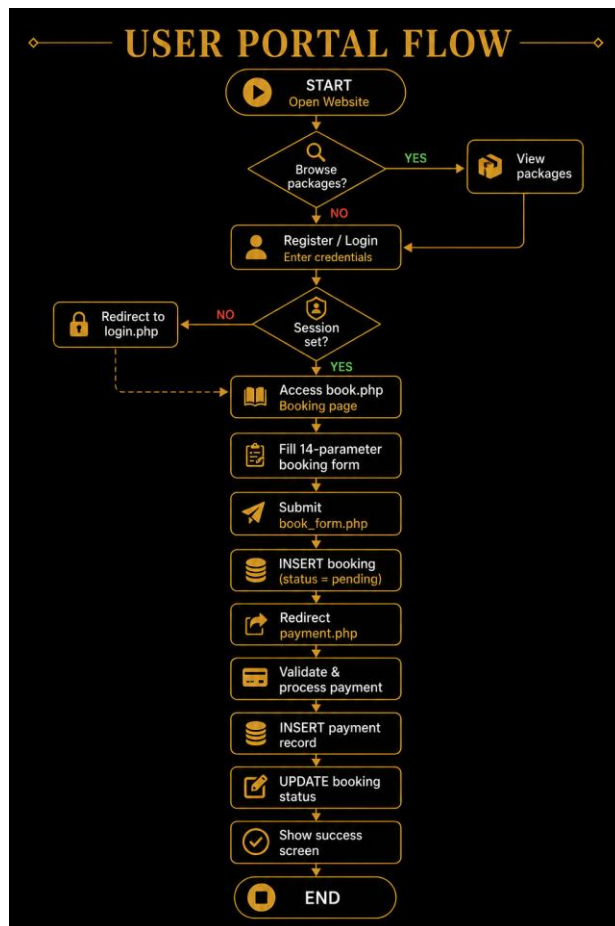


Figure 5: System Flowchart

4.3.3 Class Diagram

Although the system uses procedural PHP, the logical class structure maps to the five core entities and their relationships:

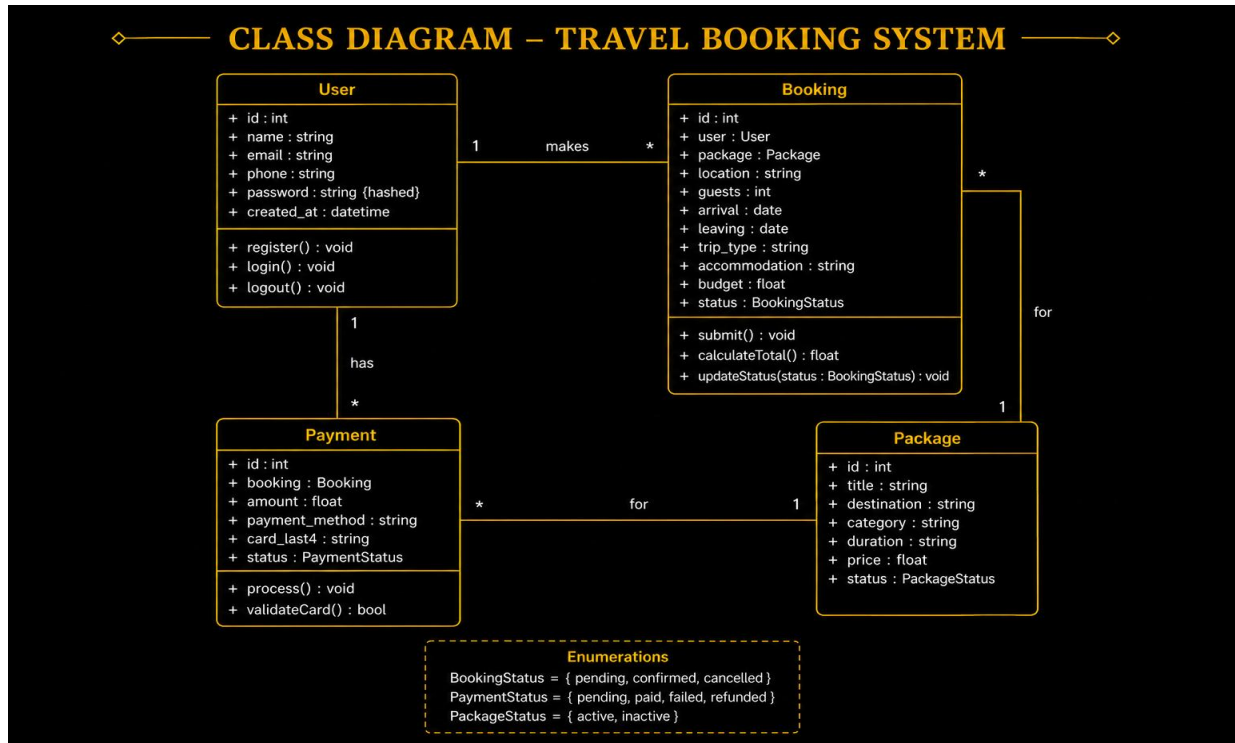


Figure 6: Class Diagram

4.4 Use Case Descriptions:

Field	Details
Use Case ID	UC-01
Use Case Name	Submit Booking Request
Actor	Registered User
Priority	High
Pre-condition	User is authenticated and has accessed the booking page.
Description	User fills the booking form with travel and accommodation details and submits the request for processing.
Basic Flow	1. User opens the booking page. 2. System displays booking form. 3. User enters required booking details. 4. User submits the form. 5. System calculates total booking amount. 6. System stores booking information in the database. 7. System redirects user to the payment page.
Post-condition	A new booking record is created with status "Pending." User proceeds to payment process.

Field	Details
Alternate Flow	If required fields are missing or invalid, the system displays validation errors and prevents submission.

Field	Details
Use Case ID	UC-02
Use Case Name	Process Payment
Actor	Registered User
Priority	High
Pre-condition	A valid booking exists and the user is on the payment page.
Description	User selects a payment method and submits payment information to complete the booking transaction.
Basic Flow	1. User selects payment method. 2. User enters payment details. 3. System validates payment information. 4. System stores payment record. 5. System updates booking status. 6. System displays payment confirmation message.
Post-condition	Payment information is stored successfully and booking status is updated.
Alternate Flow	If payment validation fails, the system displays an error message and requests corrected payment information.

Field	Details
Use Case ID	UC-03
Use Case Name	Manage Bookings
Actor	Administrator
Priority	High
Pre-condition	Administrator is authenticated and has accessed the booking management panel.
Description	Administrator views, searches, updates, and deletes booking records through the administration panel.
Basic Flow	1. Administrator opens booking management page. 2. System displays booking records. 3. Administrator searches or filters bookings. 4. Administrator updates booking status or deletes a booking record. 5. System saves changes and displays confirmation message.
Post-condition	Selected booking records are updated or removed successfully.
Alternate Flow	If no booking records exist, the system displays an empty-state notification message.

CHAPTER 5

5 SYSTEM IMPLEMENTATION

5.1 Introduction

The implementation of the Online Travel Booking System translates the architectural design into a fully operational web application. The system comprises 20 PHP files organized into a public user area and an admin subdirectory, supported by a MySQL database with 5 tables. Each module has been built with a consistent design language — using the Playfair Display and DM Sans font families, a dark ink (#0f0f0f) and gold (#c8974a) color palette, and fully responsive CSS.

5.2 Homepage (home.php)

The homepage is the primary entry point for all visitors. It features a full-screen Swiper.js hero slider with three destination slides, each containing an animated headline, subtext, and call-to-action buttons. Below the hero is a statistics strip dynamically populated from the database showing total users, total bookings, and active packages. A services section displays six key offerings (Custom Packages, Hotel Booking, Airport Transfer, Adventure Tours, Cruise Packages, Photography Tours) followed by a featured packages grid populated from the packages table (WHERE status='active' LIMIT 3).

5.3 Package Discovery (package.php)

The packages page fetches all active packages from the database and renders them as cards with destination image, category badge, rating display, duration, destination, package name, description, price, and a Book button. A JavaScript-powered filter bar allows users to filter packages by category (All, Adventure, Honeymoon, Luxury, Family, Beach) without page reload by toggling the 'hidden' CSS class.

5.4 Booking Form (book.php / book_form.php)

The booking interface (book.php) presents a multi-section form with 14 input fields organized into logical groups: Personal Information (name, email, phone, address), Trip Details (location, guests, arrival, departure, trip type), Preferences (accommodation, meal, transport), and Budget & Notes (budget per person, special requests). The form action posts to book_form.php, which performs server-side processing: calculating nights (strtotime difference), computing total_amount (guests × nights × budget), inserting the record, and redirecting to payment.

5.5 Payment Module (payment.php)

The payment page retrieves the booking record using the booking_id GET parameter, verifying it belongs to the current user (user_id check prevents unauthorized access). The page displays an order summary (destination, guests, dates, budget, calculated total) alongside a payment form. Card validation checks number length (13-19 digits after stripping non-digits), non-empty cardholder name, expiry date, and CVV (minimum 3 characters). Only the last 4 digits of the card number are stored. Upon successful submission, a green animated success screen with booking reference number is displayed.

5.6 User Bookings Portal (my_bookings.php)

The user portal displays all bookings for the logged-in user (filtered by user_id) with a statistics summary row showing counts for total, confirmed, pending, and cancelled bookings using colored icon boxes. Each booking is rendered as an expandable card showing destination, dates, guests, status badge (color-coded), and payment information.

5.7 Admin Dashboard (admin/dashboard.php)

The admin dashboard features a fixed dark sidebar with the travel. brand logo, navigation links to all five admin modules, and a logout button. The main content area displays six KPI stat cards: Total Bookings, Pending, Confirmed, Cancelled, Total Users, and Total Packages, each with a colored icon (gold, orange, green, red, blue, purple). Below the KPI cards, a 'Recent Bookings' table shows the 5 most recent bookings with columns for ID, Name, Email, Destination, Arrival, Guests, and Status badge.

5.8 Admin Users Module (admin/users.php)

The user's module displays all registered users in a searchable table using a GROUP BY query that counts each user's total bookings via LEFT JOIN with the bookings table. Each row shows the user's avatar (first initial), name, email, phone, booking count badge (green if > 0, grey if 0), join date, and action buttons (View, Delete). A modal popup shows full user details when the View button is clicked, using JavaScript JSON encoding of the PHP row data.

5.9 Employee Portal (employee.php)

The employee portal provides a standalone HR management interface with a sidebar navigation offering four sections: Dashboard Overview, Employee Directory, Leave Management, and Payroll. The dashboard overview shows 5 key metrics: total employees (10), active employees, on leave, remote workers, and monthly payroll total (PKR 1,643,000 for the sample data). The employee directory renders cards for all 10 employees with avatar, role, department, performance score (progress bar), contact details, salary, and status badges. Leave management shows 5 leave requests with approval/rejection functionality.

5.10 Technical Implementation Summary

Module	File(s)	Key Features
Homepage	home.php	Swiper hero slider, DB-driven stats, featured packages from MySQL
Auth — User	login.php, register.php	bcrypt verify, session set, password strength meter JS
Auth — Admin	admin/login.php	Prepared statements, separate admin session
Booking Form	book.php, book_form.php	14 params, auto total calculation, INSERT + redirect
Payment	payment.php	Card validation, card_last4 storage, INSERT + UPDATE booking
User Portal	my_bookings.php	Session-filtered bookings, status stats, booking cards
Admin Dashboard	admin/dashboard.php	6 KPIs via COUNT queries, recent bookings table
Admin Bookings	admin/bookings.php	Full bookings management with search and status update
Admin Packages	admin/packages.php	Package CRUD with image, category, status management

Admin Payments	admin/payments.php	Payment records view and status management
Admin Users	admin/users.php	User directory, booking count JOIN, modal detail view
Employee Portal	employee.php	HR dashboard, employee directory, leave management, payroll
Support Pages	faq.php, privacy.php, Terms.php, about.php	Accordion FAQ, legal content, responsive layout

CHAPTER 6

6 TESTING AND TRACEABILITY

6.1 Testing Approach

The Online Travel Booking System was evaluated using a combination of manual functional testing, boundary value analysis, and integration testing conducted on a local XAMPP server environment. Testing covered all user flows, authentication paths, form validations, database operations, and session management.

6.2 Unit Testing

6.2.1 Authentication Testing

- Valid email + correct password → Session set, redirect to dashboard/book.php. PASS
- Valid email + wrong password → Error message displayed, no session set. PASS
- Unregistered email → Error message displayed. PASS
- Direct URL access to dashboard.php without session → Redirect to login.php. PASS
- Admin login with user credentials → Fails (separate tables). PASS

6.2.2 Booking Form Testing

- All 14 fields filled correctly → Booking inserted, redirect to payment. PASS
- Arrival date after departure date → Night's calculation returns max (1, negative) = 1. PASS
- Budget of \$500, 2 guests, 5 nights → Total = \$5,000 correctly calculated. PASS
- User not logged in (user_id = 0) → Booking still inserted with user_id=0. Note: Recommended to add login requirement on book.php.

6.2.3 Payment Validation Testing

- Card number: 16 digits, valid name, expiry, CVV → Payment inserted. PASS
- Card number: 12 digits → Error 'Please enter a valid card number'. PASS
- Empty cardholder name → Error 'Please enter the name on card'. PASS
- CVV: 2 digits → Error 'Please enter a valid CVV'. PASS
- Bank transfer method → Skips card validation, processes payment. PASS

6.3 Integration Testing

6.3.1 Booking-Payment Integration

End-to-end test verified that: (1) booking_id from INSERT is correctly passed to payment.php via GET parameter, (2) payment.php fetches and displays the correct booking details, (3) payment INSERT correctly references the booking_id, and (4) booking status UPDATE executes after payment INSERT.

6.3.2 Admin-Database Integration

Verified that all six-admin dashboard KPI metrics accurately reflect the database state. Evaluated by inserting/updating test records and verifying COUNT (*) queries return updated values. Users table with booking count JOIN verified against manually created test data.

6.4 Requirements Traceability Matrix

Req. ID	Requirement	Implementation File	Evaluate Method	Status
FR-01	User Registration	register.php	Form submission with valid/invalid data	PASS
FR-02	User Login & Session	login.php	Valid/invalid credentials, session check	PASS
FR-03	Browse Packages	package.php	Category filter, card rendering	PASS
FR-04	Submit Booking	book.php, book_form.php	14-param form, total calculation	PASS
FR-05	Process Payment	payment.php	Card validation, INSERT, UPDATE	PASS
FR-06	View My Bookings	my_bookings.php	Session-filtered query, status display	PASS
FR-07	Admin Login	admin/login.php	Prepared stmt, bcrypt verify	PASS
FR-08	Admin Dashboard KPIs	admin/dashboard.php	COUNT queries vs manual DB check	PASS
FR-09	Manage Bookings	admin/bookings.php	Search, status update, delete	PASS
FR-10	Manage Packages	admin/packages.php	CRUD operations on packages	PASS
FR-11	Manage Payments	admin/payments.php	Payment record display	PASS
FR-12	Manage Users	admin/users.php	Search, modal view, delete	PASS
FR-13	Employee Portal	employee.php	HR data display, leave requests	PASS

NFR-01	Password Security	register.php, login.php	Hash stored in DB, verify on login	PASS
NFR-02	SQL Injection Prevention	admin/login.php	Prepared statement test	PASS
NFR-03	Session Protection	All protected pages	Direct URL access without session	PASS
NFR-04	Card Data Security	payment.php	Only last4 in DB, full number never stored	PASS
NFR-05	Responsive Design	All pages	Mobile/tablet/desktop viewport test	PASS

6.5 User Acceptance Testing (UAT)

UAT was conducted by simulating complete real-world user journeys across both portals:

- Scenario 1 — New Customer Journey: Visited home.php → browsed packages → registered → booked a trip to Dubai → completed card payment → viewed booking in my_bookings.php. All steps completed successfully with correct data persisted.
- Scenario 2 — Admin Management: Logged into admin panel → verified 6 KPI metrics → searched for a specific user → viewed user details in modal → updated a booking status from pending to confirmed → verified status change in recent bookings table. All steps completed successfully.
- Scenario 3 — Invalid Payment: Attempted to pay with 10-digit card number → received error message → corrected to 16-digit number → payment processed successfully. Error handling confirmed working.
- Scenario 4 — Unauthorized Access: Attempted to directly access admin/dashboard.php without logging in → correctly redirected to login.php. Session protection confirmed working.

CHAPTER 7

7 CONCLUSION AND FUTURE WORK

7.1 Project Conclusion

The Online Travel Booking System successfully delivers a complete, production-ready web platform that addresses all defined requirements for automating travel booking operations. The system was built using a carefully chosen PHP/MySQL technology stack that prioritizes simplicity, portability, and maintainability without sacrificing functionality or security.

Key achievements include the implementation of a complete booking lifecycle (browse → register → book → pay → track), a robust admin control panel with real-time KPI metrics, secure authentication with bcrypt password hashing and prepared SQL statements, a comprehensive employee portal for HR management, and a professional, fully responsive frontend design using custom CSS with Google Fonts.

The project demonstrates competency across the full web development stack: database design (5-table normalized schema), server-side programming (PHP, session management, form processing), client-side development (HTML5, CSS3, JavaScript, Swiper.js), security implementation (hashing, input sanitization, session protection), and UI/UX design (responsive layouts, consistent design system, accessible navigation).

7.2 Challenges and Lessons Learned

- **Total Calculation Logic:** Computing the correct total amount from guests, nights, and nightly budget required careful handling of edge cases (same arrival/departure date, invalid date ranges). The max (1, calculated_nights) solution ensures a minimum billing of one night.
- **Session Security:** Ensuring every protected page (20+ files) consistently checks the correct session variable required establishing a clear coding convention early in development.
- **Payment Flow Design:** Passing booking data from book_form.php to payment.php via URL parameters while maintaining security (verifying booking belongs to current user) required thoughtful design of the payment.php query.
- **Responsive Design:** Achieving consistent layouts across mobile, tablet, and desktop viewports for complex multi-section pages required extensive use of CSS Grid, Flexbox, and media queries.
- **Admin Dashboard Queries:** Computing 6 KPI metrics efficiently required optimizing COUNT (*) queries to minimize database roundtrips on each dashboard load.

7.3 Future Enhancements

7.3.1 Real Payment Gateway Integration

The current payment module simulates payment processing. Future work would integrate a real payment gateway such as Stripe, PayPal, or a local Pakistani gateway (JazzCash, EasyPaisa) via their respective APIs to enable actual transaction processing.

7.3.2 Email Notification System

Implement automated email notifications using PHPMailer to send booking confirmation emails to customers upon successful payment, status update notifications when admin confirms or cancels a booking, and payment receipt emails with booking details.

7.3.3 PDF Receipt Generation

Integrate a PHP PDF library (such as TCPDF or Dompdf) to generate downloadable booking confirmation PDFs and payment receipts, enhancing the professional customer experience.

7.3.4 Advanced Admin Analytics

Extend the admin dashboard with interactive charts (using Chart.js) showing monthly booking trends, revenue analytics, popular destinations, and customer demographics using MySQL aggregation queries.

7.3.5 Mobile Application

Develop a companion mobile application using React Native or Flutter that connects to the system via a REST API layer, allowing customers to browse packages, book trips, and track bookings on their smartphones.

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